



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

FLASH COLUMN VAPORIZER (E-2304)

Docu. No. : 14780-8120-PS-E2304
Project No. : 120117
Rev. No. : Z1
Date : MAY 11, 2016
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TITLE :

FLASH COLUMN VAPORIZER

(E-2304)

DISCIPLINE : **PROCESS**
DOCUMENT NO. : **14780-8120-PS-E2304**
PROJECT NAME : **PTT Phenol Train II Project**
PROJECT NO. : **120117**
OWNER : **PTT Phenol Company Limited (PPCL)**

Z1	MAY 11, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 28, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A2	APR 18, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 8, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUN 27, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 11, 2016	MAY 11, 2016	-	MAY 11, 2016	

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SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units Metric

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Case		Design	TEMA Type	BEM	Orientation	Vertical	Rev
Number of Units		1	Shells per Unit	1	Shells in Series	1	
Surface per Unit	m ²	479.1	Surface per Shell	m ²	Shells in Parallel	1	
PERFORMANCE OF ONE UNIT							
		In	SHELLSIDE	Out	In	TUBESIDE	Out
Fluid Name		S3 Steam			Cumene & CHP		
Total Flowrate (Note 2)		kg/h	7947 × 1.05		97806 × 1.05		
Vapour		kg/h	7947 × 1.05	0	0	35838 × 1.05	
Liquid		kg/h	0	7947 × 1.05	97806 × 1.05	61968 × 1.05	
Fluid Condensed		kg/h	7947 × 1.05		0		
Fluid Evaporated		kg/h	0		35838 × 1.05		
LIQUID PROPERTIES							
		AS WATER			SEE PAGES 7, 8		
Density	kg/m ³						
Viscosity	cP						
Specific Heat	kcal/kg.°C						
Thermal Conductivity	kcal/h.m.°C						
Surface Tension	dyne/cm						
Pour Point Temperature	°C						
VAPOUR PROPERTIES							
		AS WATER			SEE PAGES 7, 8		
Density	kg/m ³						
Viscosity	cP						
Specific Heat	kcal/kg.°C						
Thermal Conductivity	kcal/h.m.°C						
Hydrate Temperature	°C						
THERMAL AND HYDRAULIC DATA							
Operating Temperature	°C	135.4		130.7	70.9	97.4	
Operating Pressure	kg/cm ² g	1.78			-0.858	(Note 18)	
Number of Passes per Shell		1			1		
Average Velocity	m/s	1.40			1.75		
Pressure Drop (Allow / Calc)	kg/cm ²	Nil		0.011	0.141 (Note 8)	0.138	
Fouling Factor	h.m ² .°C/kcal	0.0001			0.0004		
Duty (Note 2)	kcal/h × 10 ⁶	4.147 × 1.05			MTD	°C	32.3
U-value Service/Clean	kcal/h.m ² .°C	311		380	Over Surface	%	10.4
		(Note 3)		(Note 3)			
DESIGN CONDITIONS							
		SHELLSIDE			TUBESIDE		
Design pressure max/min	kg/cm ² g	10.5 (Note 5)			4.0		
Full Vacuum		Required		at 195°C	Required		at 195°C
Test Pressure	kg/cm ² g	To Code			To Code		
Design Temperature max/min	°C	195		15	250		15
Corrosion Allowance	mm	3.2			1.5		
Insulation Details		Heat Conservation			Heat Conservation		
Hydrogen Service		No			No		
Sour Service		No			No		
Lethal Service		No			No		
Cyclic Service		No			No		
Cleaning Method		TBA			Mechanical		
Stress Relief for Process Reasons		No			No		
Heat Release Curve		No			Yes		

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CONSTRUCTION OF ONE SHELL

SHELL	Shell I.D.	mm	1750	Channel I.D. (If different)	mm	
TUBES	Number of Tubes		3433	Tube Length	mm	2438
	Tube O.D.	mm	19.05	Tube Thickness	mm	1.651
	Tube Gauge	BWG	16	Tube Wall Basis		Average
	Tube Pitch	mm	25.4	Tube Layout Angle	deg.	90
	Tube-Tubesheet Joint		Strength Welded (Note 9)	Tube-Baffle Diam. Clearance	mm	0.40
BAFFLES	Number of Baffles		3	Baffle Type		Single Seg.
	Baffle Cut	%	19.52	Baffle Spacing	mm	500
	Baffle Cut Orientation		Horizontal	Double Seg. Baffle Overlap	mm	0
	Baffle Thick	mm	15.9	Baffle-Shell Diam. Clearance	mm	7.9
BUNDLE	Impingement Protection		Rods (Note 10)	Tube Supports		None
	Pairs of Sealing Strips		6	Sealing Strip Details		Maximum Length
	Passlane Sealing		No	Passlane Sealing Details		N.A.
	U-Bend Diam.	mm	N.A.	U-Bend Orientation		N.A.
	U-Bend Area Included		N.A.	U-Bend Support Details		N.A.
	Longitudinal baffle		N.A.	Longitudinal Baffle Seal		N.A.

CONNECTIONS

SHELLSIDE						TUBESIDE					
Service	No.	N.B.		Rating	Facing	Service	No.	N.B.		Rating	Facing
Inlet S1	1	12	Inch	150	RFWN	Inlet T1	1	10	Inch	150	RFWN
Outlet S2	1	4	Inch	150	RFWN	Outlet T2	1	56	Inch	150	RFWN
Vent SV	1	3/4	Inch	150	RFLWN	Temp. N1	1	2	Inch	300	RFLWN
Drain SD	1	3/4	Inch	150	RFLWN	-	-	-	-	-	-
						-	-	-	-	-	-

MATERIALS

Tubes	304L Stainless Steel (Seamless)
Shell / Shell Cover	Killed Carbon Steel (UOP Std 3-11)
Floating Head Cover	N.A.
Channel / Channel Cover	304L Stainless Steel (UOP Std 3-15)
Tubesheet, Stationary	304L Stainless Steel
Tubesheet, Floating	N.A.
Baffles / Tube Supports	304 Stainless Steel
Nozzles / Nozzle Flange	Shellside: Killed Carbon Steel
Nozzles / Nozzle Flange	Tubeside: 304L Stainless Steel
Bolt and nuts	Per Project Specifications
Gaskets, Shellside	N.A.
Gaskets, Tubeside	Per Project Specifications
Gaskets, Nozzles	Per Project Specifications

ESTIMATED WEIGHTS

Weights per Shell (Tonnes)	Exchanger Weight, Dry	16.7	Exchanger Weight, Wet	27.1
	Bundle Weight, Dry	N.A.		

DESIGN REQUIREMENTS

PED Classification	Shellside- N.A.	Tubeside- N.A.
Project Specifications	UOP 963766 - 401, UOP Std 4-11-7, UOP Std 3-11, UOP Std 3-15, UOP Std 9-11-7, 14780-21A1	
Design Code	ASME VIII Div 1	
Equipment Code	TEMA Class R 9th edition, API 660 8th edition 2007	

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units

Metric

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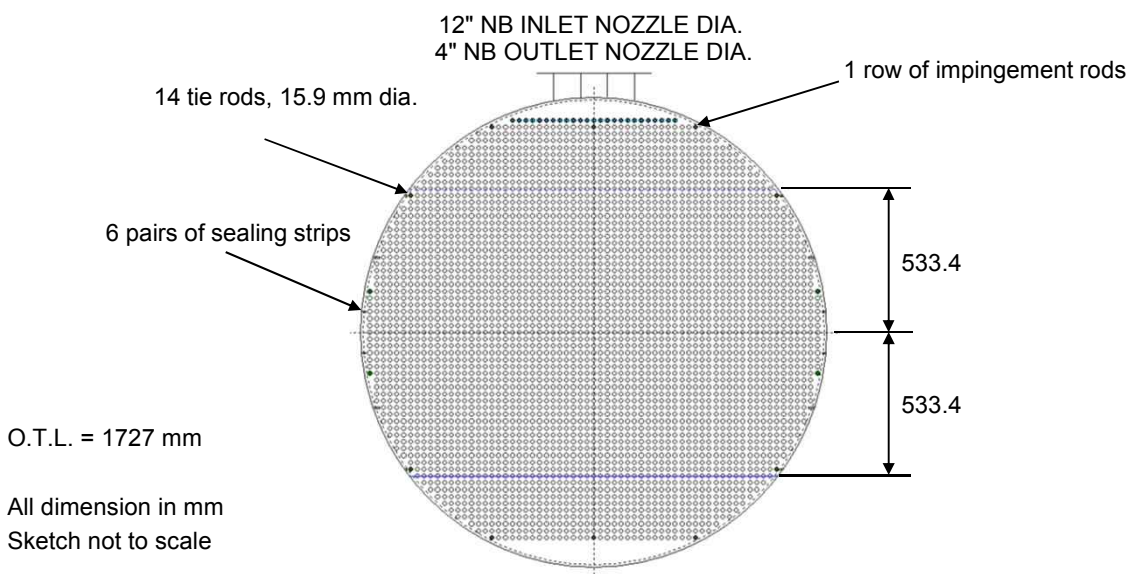
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Notes:

- Item has been checked for Flow Induced Vibration. Tube damage is not predicted.
- Flow and duty include 5% design margin.
- Overall heat transfer coefficient used in this exchanger design must not exceed 390 kcal/m²-hr-°C. VENDOR to confirm.
- Provide an axial outlet with elbow to match feed nozzle on flash column V-2302. See UOP Project Specification 963766-301 (vessels) for additional details.
- Reliable cooling water and fire water may be used as an emergency coolant through the shell side. (Design pressure to be confirmed by CONTRACTOR).
- Alternate design steam conditions are 2.8 kg/cm²g at 150°C. Saturation temperature is 142°C.
- Provide an expansion joint in the shell. Expansion joint design shall consider start-up conditions where the liquid inside the tubes is at 15°C and the shellside medium is at the maximum design temperature. See page 9.
- The allowable pressure drop on the tubeside includes static head.
- Strength-weld tubes to tubesheet per requirements on page 10.
- Provide 1 row of impingement rods of the same diameter, same pitch, and same layout as the tubes.
- Deleted
- Provide a NPS 2" 150#RF nozzle on outlet elbow so that a thermocouple may be inserted directly into one tube. Refer to UOP 963766-640.
- Provide a NPS 3/4" flanged vent connection for non-condensable gases located near the top of the shell on the side opposite the shellside inlet.
- Provide a NPS 3/4" drain connection located near the bottom of the shell on the opposite side of the shell outlet.
- The tubeside inlet nozzle shall be axial to drain completely.
- Based on preliminary check of the existing unit 1 exchanger design (refer to PPCL data sheet no. 13850-8100-DS-0042; E-1304);
The existing performs -30.7% undersurface at new duty of 4.147 x 1.05 MMkcal/h.
The existing design is not thermally suitable to reuse with the new condition.
CONTRACTOR to confirm.
- Tubesheet vent/drain should be required. CONTRACTOR to confirm.
- Based on the properties curves extrapolation for the tubeside outlet temperature of 97.4°C. Refer to UOP 963766-401, the specified inlet pressure is -0.8494 kg/cm²g.
- Deleted.

Tube Layout



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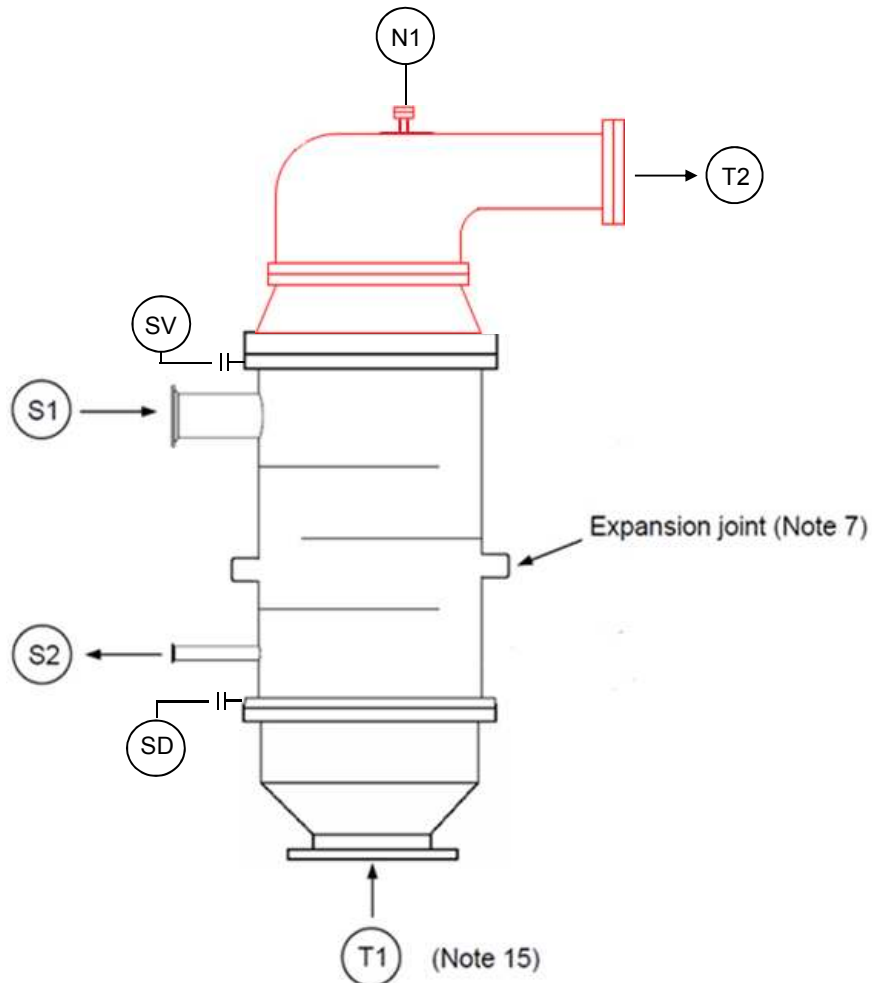
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Setting Plan



Shell ID : 1750 mm

Tube length : 2438 mm

PROCESS NOZZLE SUMMARY

REF	SERVICE	SIZE	RATING	ELEV.	REMARK
S1	S3 Steam Inlet	12"	150#RFLWN		
S2	S3 Steam Outlet	4"	150#RFLWN		
T1	Cumene & CHP Inlet	10"	150#RFLWN		
T2	Cumene & CHP Outlet	56"	150#RFLWN		
N1	Thermo Couple Connection	2"	300#RFLWN		
SV	Shellside Vent	3/4"	150#RFLWN		CW/blind
SD	Shellside Drain	3/4"	150#RFLWN		CW/blind
-	-	-	-		-
-	-	-	-		-

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ColdSide Property Curve 1

Pressure

-0.848

kg/cm²g

Temperature	Entire Stream	Entire Stream	Vapour	Vapour	Vapour	Vapour
°C	Enthalpy	Mass Vapour	Density	Viscosity	Therm. Cond.	Spec. Heat
	kcal/kg	fraction	kg/m ³	cP	kcal/h.m.°C	kcal/kg.°C
65.2	0.00	0.0000	-	-	-	-
78.8	5.13	0.0000	-	-	-	-
91.9	10.27	0.0000	-	-	-	-
104.6	15.40	0.0000	-	-	-	-
116.9	20.53	0.0000	0.6824	0.008	0.0143	0.38
118.4	25.75	0.0551	0.6840	0.008	0.0145	0.38
119.8	30.97	0.1091	0.6856	0.008	0.0147	0.38
121.3	36.18	0.1622	0.6872	0.008	0.0149	0.38
122.7	41.40	0.2144	0.6888	0.009	0.0150	0.39
124.0	46.61	0.2658	0.6920	0.009	0.0152	0.39

Temperature	Liquid	Liquid	Liquid	Liquid	Liquid	Critical	Critical
°C	Density	Viscosity	Therm. Cond.	Spec. Heat	Surf. Tension	Pressure	Temperature
	kg/m ³	cP	kcal/h.m.°C	kcal/kg.°C	dyne/cm	kg/cm ² [a]	°C
65.2	920.6	1.663	0.081	0.37	22.9	33.5	345
78.8	907.2	1.297	0.079	0.39	21.6	33.5	345
91.9	894.0	1.040	0.077	0.40	20.3	33.5	345
104.6	880.9	0.852	0.074	0.41	19.1	33.5	345
116.9	867.9	0.712	0.073	0.42	17.9	33.5	345
118.4	870.7	0.727	0.072	0.42	17.8	33.6	344
119.8	873.4	0.741	0.071	0.42	17.6	33.6	344
121.3	876.0	0.755	0.070	0.41	17.5	33.6	343
122.7	878.4	0.766	0.070	0.41	17.3	33.7	342
124.0	880.6	0.777	0.069	0.41	17.2	33.7	342

ColdSide Property Curve 2

Pressure

-0.927

kg/cm²g

Temperature	Entire Stream	Entire Stream	Vapour	Vapour	Vapour	Vapour
°C	Enthalpy	Mass Vapour	Density	Viscosity	Therm. Cond.	Spec. Heat
	kcal/kg	fraction	kg/m ³	cP	kcal/h.m.°C	kcal/kg.°C
65.2	0.00	0.0000	-	-	-	-
77.7	4.74	0.0000	-	-	-	-
89.9	9.47	0.0000	-	-	-	-
101.7	14.21	0.0000	0.3973	0.008	0.0129	0.37
103.5	19.61	0.0559	0.3973	0.008	0.0131	0.37
105.5	25.01	0.1105	0.3973	0.008	0.0133	0.37
107.5	30.41	0.1637	0.3973	0.008	0.0136	0.37
109.5	35.81	0.2158	0.3973	0.008	0.0138	0.37
111.5	41.21	0.2670	0.3989	0.008	0.0140	0.38
113.3	46.61	0.3174	0.4005	0.008	0.0143	0.38

Temperature	Liquid	Liquid	Liquid	Liquid	Liquid	Critical	Critical
°C	Density	Viscosity	Therm. Cond.	Spec. Heat	Surf. Tension	Pressure	Temperature
	kg/m ³	cP	kcal/h.m.°C	kcal/kg.°C	dyne/cm	kg/cm ² [a]	°C
65.2	920.6	1.663	0.081	0.37	22.9	33.5	345
77.7	908.2	1.321	0.079	0.38	21.7	33.5	345
89.9	896.0	1.074	0.077	0.40	20.5	33.5	345
101.7	883.9	0.891	0.075	0.41	19.4	33.5	345
103.5	887.2	0.915	0.074	0.40	19.2	33.6	344
105.5	890.4	0.937	0.073	0.40	19.0	33.6	343
107.5	893.4	0.957	0.072	0.40	18.7	33.7	343
109.5	896.2	0.974	0.071	0.40	18.5	33.7	342
111.5	898.6	0.987	0.071	0.40	18.3	33.8	341
113.3	900.6	0.998	0.070	0.39	18.1	33.8	341

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ColdSide Property Curve 3

Pressure

-0.999

kg/cm²g

Temperature	Entire Stream	Entire Stream	Vapour	Vapour	Vapour	Vapour
°C	Enthalpy	Mass Vapour	Density	Viscosity	Therm. Cond.	Spec. Heat
	kcal/kg	fraction	kg/m ³	cP	kcal/h.m.°C	kcal/kg.°C
65.2	0.00	0.0000	-	-	-	-
68.8	1.33	0.0000	-	-	-	-
72.3	2.66	0.0000	0.1281	0.007	0.0103	0.34
74.6	8.94	0.0637	0.1281	0.007	0.0105	0.34
77.4	15.22	0.1252	0.1281	0.007	0.0108	0.35
80.8	21.50	0.1839	0.1265	0.007	0.0111	0.35
84.5	27.78	0.2400	0.1265	0.008	0.0115	0.35
88.3	34.06	0.2938	0.1265	0.008	0.0119	0.36
91.8	40.34	0.3466	0.1265	0.008	0.0123	0.36
94.8	46.61	0.3995	0.1265	0.008	0.0127	0.36

Temperature	Liquid	Liquid	Liquid	Liquid	Liquid	Critical	Critical
°C	Density	Viscosity	Therm. Cond.	Spec. Heat	Surf. Tension	Pressure	Temperature
	kg/m ³	cP	kcal/h.m.°C	kcal/kg.°C	dyne/cm	kg/cm ² [a]	°C
65.2	920.6	1.663	0.081	0.37	22.9	33.5	345
68.8	917.1	1.555	0.080	0.38	22.6	33.5	345
72.3	913.6	1.457	0.080	0.38	22.2	33.5	345
74.6	918.4	1.527	0.078	0.38	22.0	33.6	344
77.4	923.1	1.592	0.077	0.37	21.6	33.7	343
80.8	927.5	1.645	0.076	0.37	21.2	33.7	342
84.5	931.2	1.678	0.074	0.37	20.8	33.8	341
88.3	933.9	1.687	0.073	0.37	20.3	33.8	339
91.8	935.6	1.679	0.072	0.37	19.9	33.9	339
94.8	936.5	1.663	0.071	0.37	19.5	33.9	338

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MEAN METAL TEMPERATURES TO BE USED FOR FIXED TUBESHEET DESIGN

Mean Metal Temperature for all potential cases to be confirmed by CONTRACTOR.

TBA = To Be Advised by CONTRACTOR

DESCRIPTION OF CASE	NUMBER OF CYCLES (MAX)	SHELL		TUBE	
		MEAN METAL TEMP	OPERATING PRESSURE	MEAN METAL TEMP	OPERATING PRESSURE
		°C	kg/cm²g	°C	kg/cm²g
NORMAL OPERATING CONDITIONS (FOULED)	1000	130.8	1.762	126.0	-0.8568
START UP CONDITIONS (Note 7)	1000	TBA	1.762	TBA	-0.8568
SHUT DOWN CONDITIONS		N.A.	N.A.	N.A.	N.A.
ZERO FLOW ON SHELLSIDE		Ambient	FV	TBA	TBA
ZERO FLOW ON TUBESIDE		TBA	TBA	TBA	FV
EMERGENCY OR UPSET CONDITIONS		TBA	TBA	TBA	TBA
STEAM OUT BOTH SIDE SIMULTANEOUSLY		TBA	TBA	TBA	TBA
STEAM OUT ON SHELLSIDE ONLY		N.A.	N.A.	N.A.	N.A.
STEAM OUT ON TUBESIDE ONLY		TBA	TBA	TBA	TBA
DESIGN PRESSURE (kg/cm²g)		10.5 / FV		4.0 / FV	
DESIGN TEMPERATURE (°C)		195 / 15		250 / 15	
HYDRAULIC PRESSURE TEST:					
SHELLSIDE		Ambient	CODE	Ambient	ATM
TUBESIDE		Ambient	ATM	Ambient	CODE

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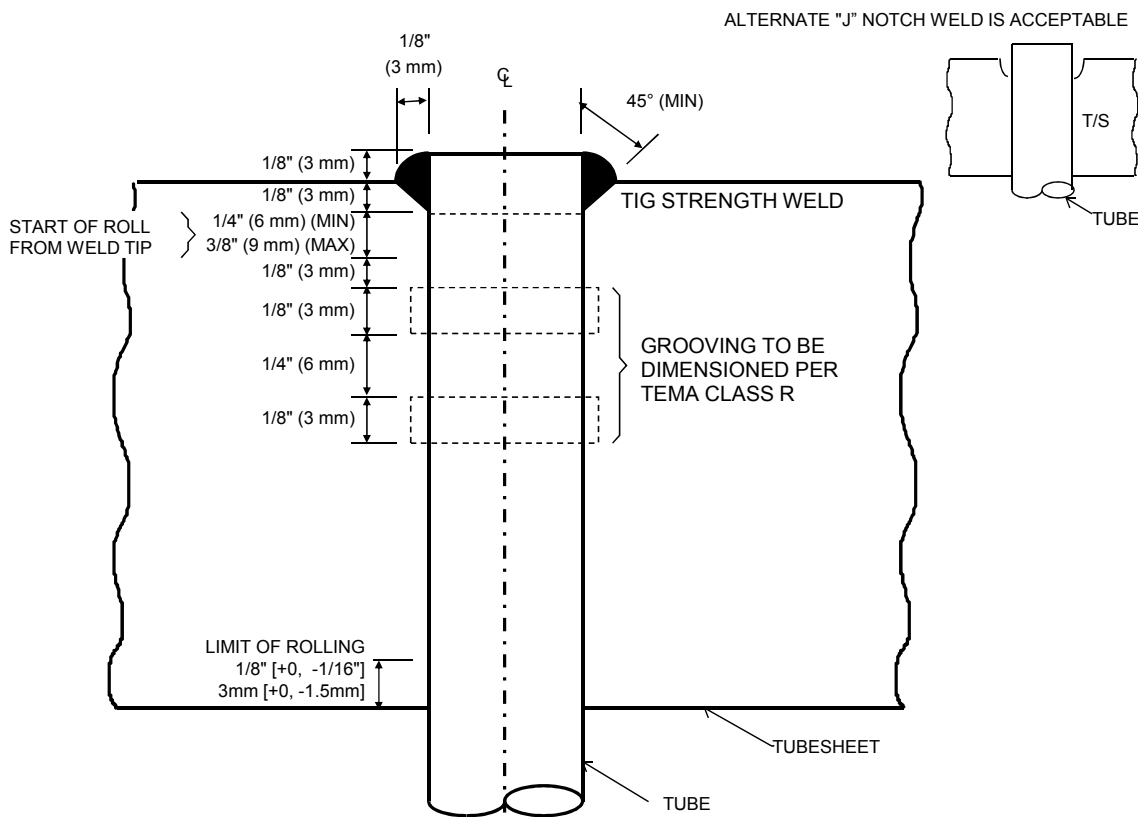
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Strength-Weld Type Tube to Tubesheet Joint Requirements (Refer to UOP 963766 - 401)



1. Tubing shall be seamless type.
2. Joint components shall be deburred and meticulously cleaned prior to welding.
3. When heat treatment of the tubesheet is required, final tube hole sizing shall be attained by reaming after heat treatment.
4. Preheat tubesheet to a minimum of 150°F (65°C) and maintain this temperature during welding.
5. Each tube end shall be welded individually without stopping.
6. Tube to tubesheet joints shall be welded with a minimum of two passes. The first pass shall be 100% examined by dye penetrant or magnetic particle method. Remaining passes shall be welded without interruption. The final welds shall be 100% examined by dye penetrant or magnetic particle method.
7. All tube joint welds shall be proven using a soap test [with minimum 15 psi (1.05 kg/cm²) air] prior to rolling.
8. A helium leak test shall be performed in accordance with the requirements of ASME code, Section V, Article 10. The shell side shall be pressurized with helium while the exchanger is inclined to provide good flow of helium to a single collection point on the tube side. Any leakage in excess of 1×10^{-4} standard cc/sec warrants repair.
9. Do all tube rolling after completion of welding and leak testing.
10. The tubes shall be rolled for the entire tubesheet thickness. Stop shall be used on tube roller that assures expansion is within limits given on sketch.
11. A certified test report of the final, successful helium test shall be provided and signed by an independent third party witness.
12. All applicable Weld Procedure Specifications (WPS) and Welder Qualification Records (WQR) shall be provided per clients request.

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